

Panel MS-AR(1): common linear trend, $\rho = 0.99$

Seasonally adjusted real GDP per worker

1 Model

One specification. Joint panel Markov-switching AR(1) around a common linear trend. No country-specific random effects. Outcome is log seasonally adjusted real GDP per worker (2026-09-17 vintage), full sample.

$$y_{it} = a + g t + z_{it}, \quad (1)$$

$$z_{i,t+1} = (1 - \rho)\mu(s_{it}) + \rho z_{it} + \sigma(s_{it}) \varepsilon_{it}, \quad (2)$$

$$s_{i,t+1} \sim \Pi(\cdot \mid s_{it}). \quad (3)$$

Three regimes. $\rho = 0.99$ is fixed and common to every regime. Median μ pinned at 0. σ switches. Standard errors are delta-method from a numerical Hessian (not reported for the fixed ρ).

2 Common intercept and linear trend, rho fixed at 0.99

2026-09-17 SA panel, full sample, $y_{it} = a + gt + z_{it}$, $\rho = 0.99$ fixed. 74 countries, 8670 observations, 1950-Q1–2026-Q2. Fitted countries: 74. Observations: 8670. Log-likelihood: 22607.21. Ergodic mean of the cycle $E[z] = 0.7046$ (0.0694). Common intercept $a = -6.0152$ (0.0594). Common slope $g = 0.0080$ (0.0005).

Table 1: Regime AR(1), transition matrix, and ergodic π . Columns are regimes.

	(0)	(1)	(2)
μ	-0.2514 (0.0953)	0.0000 —	1.7443 (0.0463)
σ	0.0699 (0.0020)	0.0161 (0.0004)	0.0090 (0.0002)
ρ (common)	0.9900	0.9900	0.9900
$\Pi_0.$	0.7351 (0.0241)	0.1931 (0.0222)	0.0718 (0.0113)
$\Pi_1.$	0.0650 (0.0066)	0.9300 (0.0067)	0.0050 (0.0022)
$\Pi_2.$	0.0280 (0.0034)	0.0023 (0.0015)	0.9698 (0.0033)
π	0.1492 (0.0119)	0.4253 (0.0305)	0.4255 (0.0323)

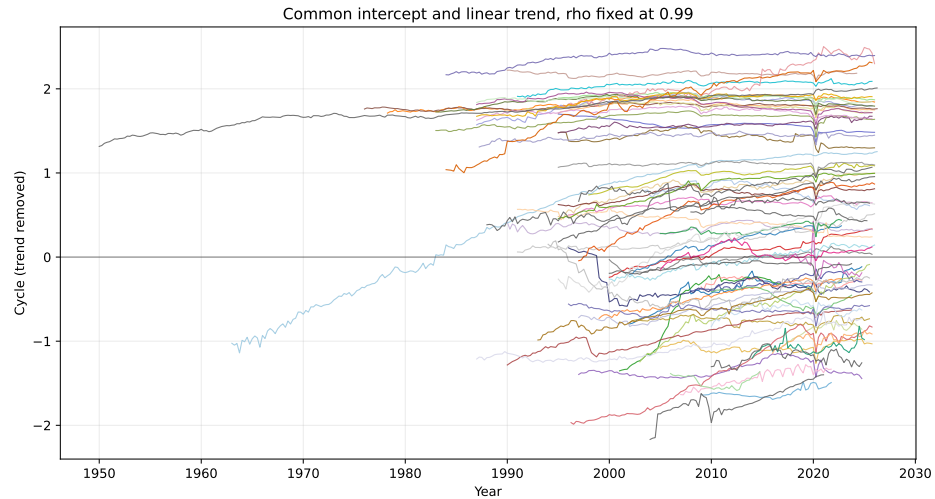


Figure 1: Country cycles after removing the common intercept and linear trend.